

Integrating MUSLE and sediment trap efficiency for oxbow lake service-life estimation, Krueng Keureuto Watershed

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Abstract: Annual flooding in the Krueng Keureuto Watershed causes significant losses to the surrounding community. One flood-control measure is the use of an oxbow lake, which is cyclical and lake that stores excess water when river discharge increases, provides clean water, and supports irrigation. However, the lake's main problem is sedimentation, which can reduce its capacity and shorten its service life. Therefore, this study analyse the sedimentation rate and estimate the lake's service life. The sedimentation analysis was conducted using the modified universal soil loss equation (MUSLE) method. The study used rainfall data from 2011 to 2023, a topographic map generated from digital elevation model (DEM) data, soil-type and land-use data, and lake storage capacity. The analysis shows that the oxbow lake catchment area is dominated by inceptisols, while primary dryland forest is the largest land-use category, covering 276.12 km². The slope length and steepness (*LS*) factor value was 2.33. With an annual sedimentation rate of 53.47 m³·y⁻¹, the oxbow lake has an operational lifespan of 3.9 years, at which point the dead storage capacity is filled with sediment. These results can support management in determining maintenance strategies, dredging plans, and other measures to sustain the lake's function for flood control and downstream water supply.

Keywords: dead storage, lake lifespan, modified universal soil loss equation (MUSLE), oxbow lake, sedimentation

INTRODUCTION

Sedimentation is the process of material deposition due to erosion caused by water flow, which can reduce the storage capacity of reservoirs and lakes (Armido *et al.*, 2020; Basri *et al.*, 2020; Ziana *et al.*, 2022; Ajaaj *et al.*, 2024; Azmeri *et al.*, 2025a; Azmeri *et al.*, 2025b). Sedimentary material, consisting of sand and other particles, is transported by erosion processes and ultimately settles on the riverbed (Maini *et al.*, 2024). The main factors driving sedimentation are high rainfall and human activities, such as land use changes. This problem also occurs in Krueng Keureuto, Aceh, which plays a vital role in providing irrigation and raw water (Dinas Pengairan, 2022), but is frequently hit by annual flooding (Azri, 2023; CNN, 2023).

Efforts to control flooding have been implemented, including the construction of the Keureuto Reservoir (Dinas Pengairan, 2022); however, flooding continues. In 2024, an oxbow lake was

designed to reduce flooding (Harahap, 2025; Istiqfarah, 2025). An oxbow lake is formed from a meander that is cut off from the main river flow (Ishii and Hori, 2016). It is expected to retain excess river discharge to prevent it from flowing directly downstream.

An oxbow lake is a shallow lake or pond that forms naturally on a floodplain when a river meanders and the bend is interrupted by erosion and sedimentation. It then forms an isolated body of water alongside the main river. Oxbow lakes typically exist when connectivity to the main river is limited, but they can reconnect during major floods. Over time, they undergo progressive sedimentation and transform into wetlands or swamps (Cantonati *et al.*, 2020). Based on the flood management master plan (Dinas Pengairan, 2022) and a previous study (Harahap, 2025), oxbow lakes can be optimised for flood mitigation during high river flows and for providing clean water and irrigation during the dry season. Most importantly, oxbow lakes limit land use because they are located in river areas, thereby

avoiding land-acquisition costs. However, the negative impacts of sedimentation on oxbow lakes are quite significant (Guo *et al.*, 2023; Ajaaj *et al.*, 2024). It is described below.

1. Reduced storage capacity and flood control effectiveness: Sedimentation at the bottom of the lake may reduce the lake's water-holding volume. This condition reduces the lake's effectiveness by shortening its lifespan, thereby affecting its ability to meet water needs and reduce flooding, increasing the risk of overflows, and potentially flooding surrounding areas.
2. Changes in the lake's flow patterns and morphology: sediment accumulation can alter the lakebed (topography), causing uneven water flow and increasing the risk of local shallowing (shallow zones), which may interfere with its operation.
3. Disruption to water quality: sediment often carries organic material, heavy metals, and other pollutants that can degrade lake water quality, thereby affecting its suitability for domestic and agricultural use.
4. Increased maintenance frequency and costs: lakes experiencing high sedimentation require periodic dredging to restore their storage capacity. A budget must be regularly allocated for maintenance.

This research is essential for examining the effect of sedimentation rate on the operational life of the Krueng Keureuto oxbow lake. Krueng Keureuto Watershed, a national strategic watershed, was selected as the study location due to its recurring flooding. The Keureuto Watershed has complete river cross-section and morphological data, including data on oxbows, hydrology, hydraulics, and land cover. These data support the development of a valid model. This study employed the MUSLE method to estimate sediment volume based on peak discharge and annual runoff volume, ensuring the estimated sedimentation rate reflects actual conditions each year. In addition, trap efficiency analysis was used to determine the lake's operational life in retaining incoming sediment. The combination of sedimentation rate calculations using MUSLE and trap efficiency assessments enables a more accurate determination of the lake's operational life.

Despite extensive sedimentation studies in conventional reservoirs, limited research has examined sediment dynamics and operational lifespan in natural oxbow lakes, particularly in tropical floodplain environments where hydrological connectivity with the main river is strongly influenced by flood events. Unlike reservoirs with relatively stable inflow conditions, oxbow lakes experience cyclic connectivity that significantly affects sediment inflow and deposition processes. Therefore, this study contributes by applying an integrated approach that combines MUSLE-based sediment yield estimation with trap efficiency analysis to evaluate the operational lifespan of an oxbow lake in a flood-prone tropical watershed. This approach provides new insight into sediment accumulation and the short-term sustainability of oxbow lakes as nature-based flood mitigation systems under extreme tropical rainfall regimes.

MATERIALS AND METHODS

STUDY LOCATION

The Krueng Keureuto study area has a total watershed area of 932.11 km² and an oxbow lake catchment area of 604.91 km². The upstream area is located between coordinates 4°45'18" N and

97°9'26.96" E, and the downstream area is located between coordinates 5°2'0.83" N and 97°19'20.72" (Fig. 1).

In oxbow lake systems, hydrological connectivity between the main river and the oxbow lake generally occurs during flood events. When river discharge exceeds a certain threshold, water and sediment are transported into the oxbow lake through the connecting channel. During normal flow conditions, the oxbow lake remains relatively isolated from the main river, resulting in limited sediment exchange. This periodic connectivity plays an important role in controlling sediment inflow and accumulation within the oxbow lake.

RESEARCH DATA AND EQUIPMENT

The data and equipment required were as follows.

1. Rainfall data: maximum annual rainfall from 2011 to 2023 from three stations located around the Krueng Keureuto Watershed, including ARR Jambo Aye, ARR Krueng Keureuto, and ARR Panton Labu. The data were processed using the Soil Conservation Service (SCS) method to obtain peak discharge (Q_p) and runoff volume (V_q).
2. A topographic map of the watershed was generated from a Digital Elevation Model (DEM) processed using ArcGIS 10.8. This map determined the oxbow watershed boundaries, as well as slope length and steepness, which are factors in the LS.
3. A map of soil types obtained from the Dinas Pengairan (2022) containing information on soil texture, structure, and permeability. This map was used to determine soil erodibility factors.
4. A map of land use obtained from the Dinas Pengairan (2022). This map was used to determine land cover management factors and soil conservation measures.
5. Software: ArcGIS 10.8 was used for data integration, spatial analysis, and visualisation. Image processing was performed using ArcGIS 10.8 and Google Earth.
6. The operational life of the oxbow lake was determined using the trap efficiency method, which measures the percentage of sediment retained by the oxbow lake relative to the total incoming sediment.

STUDY PROCEDURE

The complete flowchart is shown in the supplementary material (Fig. S1). The research was conducted using ArcGIS 10.8 to analyse data from all MUSLE parameters using the equations developed by Williams (1975) as cited in Azmeri *et al.* (2024, p. 3). Sediment yield (SY , in Mg·y⁻¹) is calculated using Equation (1):

$$SY = a(V_q \cdot Q_p)^b \cdot K \cdot LS \cdot CP \quad (1)$$

where: V_q = flow volume (m³), a = coefficient = 11.80, b = coefficient = 0.56, K = soil erodibility factor, LS = slope length and steepness, C = land cover type factor, P = land management factor.

Peak flood discharge is the maximum river flow rate that occurs during a rainfall event with a specific return period. This study used the Soil Conservation Service (SCS) method with a dimensionless unit hydrograph. The important components of the SCS unit hydrograph are the basin lag time and peak discharge (Q_p), as shown in the Equation (2) (Kurniawan *et al.*, 2022).

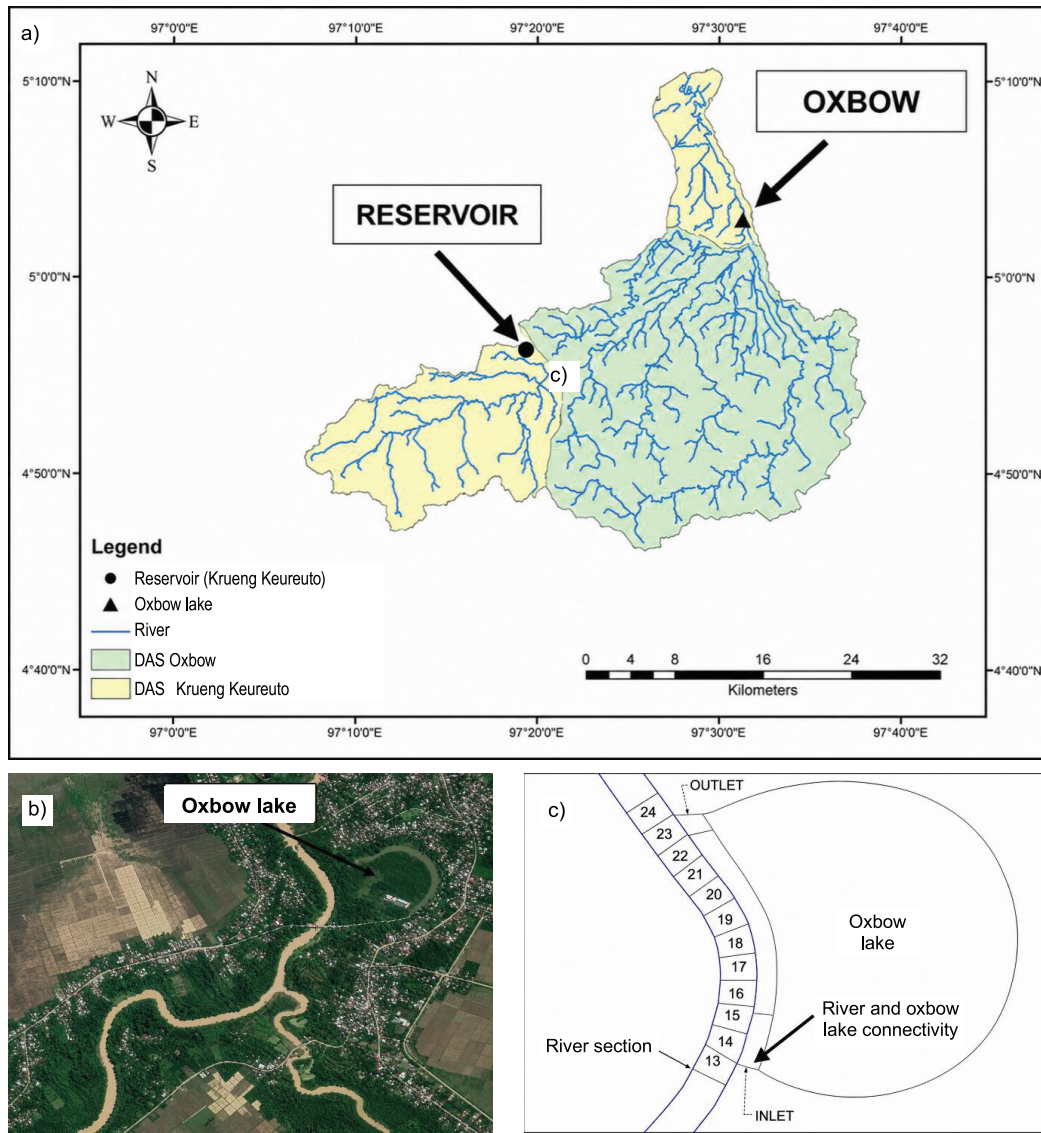


Fig. 1. Study location: a) Krueng Keureuto Watershed, b) the oxbow lake and c) conceptual connectivity between river and oxbow lake during flood events; source: own elaboration based on Harahap (2025)

$$Q_p = \frac{0.208A}{t_p} \quad (2)$$

where: Q_p = flood discharge ($\text{m}^3 \cdot \text{s}^{-1}$), A = catchment area (km^2), t_p = time to peak (h).

Surface runoff is the flow of rainwater that falls in upland areas and moves toward lower areas. Runoff volume was calculated using the curve number (CN), which estimates runoff depth from rainfall intensity during a given rainfall event according to Equation (3) (Chin, 2023).

$$V_q = V_a \cdot A \quad (3)$$

where: V_q = flow volume during an event (m^3), V_a = annual runoff volume value (mm).

The soil erodibility factor (K) reflects the soil's susceptibility to erosion due to rainfall and surface runoff, influenced by soil texture, structure, organic content, and permeability (Komissarov *et al.*, 2024). The slope length and steepness (LS) factor describes the combined effect of slope length and steepness on erosion severity. Longer and steeper slopes increase the potential for soil

transport (Azmeri *et al.*, 2022; Alemu *et al.*, 2025; Azmeri *et al.*, 2025c). Land use and crop management (C) factors determine soil loss estimates because they are directly related to land use and land cover (Tsai *et al.*, 2021). Meanwhile, the conservation action factor (P) is influenced by slope gradient and the conservation methods applied (Tsige, Malcherek and Saleshi, 2022). Spatial analysis using GIS enables accurate characterisation of watershed parameters and improves the estimation of erosion-related factors required for soil erosion modelling (Azmeri, Legowo and Rezkyna, 2020).

The sediment delivery ratio (SDR , %) is the ratio between sediment reaching a river and total soil loss due to erosion (Xu *et al.*, 2024). The SDR provides an overview of the proportion of upstream sediment that actually reaches the river (Irawan *et al.*, 2025).

$$SDR = S \frac{1 - 0.8683(A^{-0.2018})}{2 + S + 50n} + 0.8683A^{-0.2018} \quad (4)$$

where: S = slope (%), n = surface roughness coefficient (Manning).

The amount of sediment carried to the Krueng Keureuto River (SY_{SDR} , $\text{Mg}\cdot\text{y}^{-1}$) was calculated using the following formula.

$$SY_{SDR} = SDR \cdot SY \quad (5)$$

Sediment inflow analysis was performed to determine the amount of sediment actually entering the oxbow lake. The dynamics of sediment supply to the oxbow are influenced by hydrological connectivity and flood intensity; therefore, not all sediment from land erosion enters the water body (Downs and Piegay, 2019). The amount of sediment carried to the oxbow lake ($\text{Mg}\cdot\text{y}^{-1}$) is calculated as in the Equation (6):

$$V_s = \frac{\sum Q_{\text{inflow}}}{\sum Q_{\text{flood}}} SY_{SDR} \quad (6)$$

where: V_s = amount of sediment carried to the oxbow lake ($\text{Mg}\cdot\text{y}^{-1}$), Q_{inflow} = inflow discharge to the oxbow lake ($\text{m}^3\cdot\text{s}^{-1}$), Q_{flood} = flood inflow to the oxbow lake ($\text{m}^3\cdot\text{s}^{-1}$).

Values of Q_{inflow} and Q_{flood} used in the Equation (6) were obtained from hydrological analyses reported by Harahap (2025). The inflow discharge represents the normal flow entering the oxbow lake, while the flood discharge reflects higher flow conditions during flood events that transport significant amounts of sediment into the lake.

The oxbow lake lifespan analysis is used to estimate the extent to which the lake's storage capacity can withstand before silting occurs due to sediment accumulation. The lake's lifespan is calculated using trap efficiency, which indicates the percentage of sediment retained by the lake compared to the total incoming sediment (Septianita *et al.*, 2020). Trap efficiency is calculated using the Brune method, which uses an empirical curve to describe the relationship between trap efficiency and the ratio of storage capacity to inflow (C/I) (Mulu and Dwarakish, 2015). Trap efficiency was calculated using the Brune method by reading a graph based on the lake's storage capacity and annual inflow, as shown in Figure S4. The sediment yield retained in the oxbow lake was initially expressed as sediment mass ($\text{Mg}\cdot\text{y}^{-1}$). To estimate the sediment volume required for the lifespan analysis, the retained sediment mass was converted to sediment volume ($\text{m}^3\cdot\text{y}^{-1}$) using the bulk density of the deposited sediment. The bulk density value of 2.56 was obtained from primary field measurements conducted in the study area.

The lifespan of the oxbow lake T (year) can be calculated using the Equation (7).

$$T = \frac{V}{V_s \cdot \eta} \quad (7)$$

where: V = dead storage capacity of the oxbow lake (m^3), V_s = amount of sediment carried to the oxbow lake ($\text{m}^3\cdot\text{y}^{-1}$), η = trap efficiency (%).

RESULTS

HYDROLOGY ANALYSIS

Average rainfall data from three stations, calculated using the Thiessen Polygon method, are provided in the supplementary material (Tab. S1). Furthermore, the alternating block method

(ABM) was applied to examine the rainfall duration of seven hours during the period from 2011 to 2023 (Tab. S2). The determined rainfall duration, adjusted to local rainfall event characteristics, is considered appropriate because it reflects the watershed's response time to extreme rainfall events.

The planned rainfall distribution using the excess rainfall hyetograph (ERH) method for 2011–2023 is presented in Table S3. The planned rainfall is divided into specific time intervals, resulting in a more detailed rainfall-intensity distribution. The curve number (CN) was calculated in ArcGIS 10.8 using soil type and land-use factors. The obtained CN value was 69 and was used to calculate effective rainfall.

The results of the planned rainfall distribution were used as inputs in the Soil Conservation Service (SCS) method. Using the SCS method, the effective rainfall is calculated based on soil characteristics and land use. Subsequently, the runoff volume (V_q) was defined as the total runoff generated during each rainfall event, and the peak discharge (Q_p) was defined as the maximum discharge during the runoff period. Both parameters relate to effective rainfall, land infiltration capacity, and hydrograph unit characteristics. The results of the SCS method calculation are provided in full in Table S4 and Figure S2, while the peak discharge (Q_p) values are provided in Table S5 and Figure S3. The surface runoff volume values (V_q) are provided in Table S6.

SOIL ERODIBILITY FACTOR (K)

The soil erodibility factor (K) is the degree of soil sensitivity to erosion, determined by soil physical properties such as texture, structure, organic matter, and permeability. The soil type at the location is shown in Figure 2, and the K factor values are shown in Table 1.

SLOPE LENGTH AND STEEPNESS (LS)

The LS factor, obtained using the ArcGIS 10.8 software, is presented in Figure 3.

The topographic analysis resulted in a slope length of 26.69 m and a slope angle of 15.43%. The slope length factor (L) was 1.12, and the slope gradient factor (S) was 2.08. Thus, the LS value was 2.33.

CROP AND LAND CONSERVATION MANAGEMENT FACTOR (CP)

The CP factor describes the extent to which agricultural and land conservation activities affect the degree of erosion, as shown in Figure 4 and Table 2.

SEDIMENTATION RATE AT THE KRUENG KEUREUTO OXBOW LAKE CATCHMENT AREA

Sedimentation at the Krueng Keureuto oxbow lake catchment area was determined based on peak discharge, surface runoff volume, soil erodibility, land-cover conditions, and slope-gradient data. The results are presented in Table 3 and Figure 5. The distribution of each sedimentation class was analysed to determine the level of sedimentation occurring in the study area (Tab. 4).

A sediment delivery ratio (SDR) analysis was conducted to determine the amount of sediment that actually reaches the

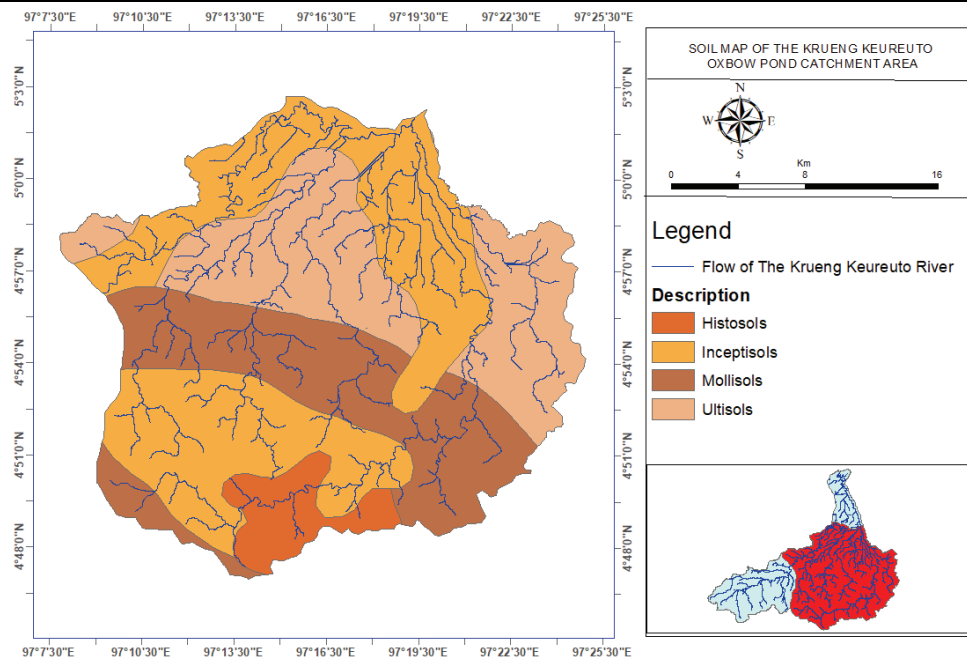


Fig. 2. Map of soil types at the Krueng Keureuto oxbow lake catchment area; source: own study

Table 1. Soil types, area (A), soil erodibility factor (K), percentage, and composite K value of the Krueng Keureuto oxbow lake catchment area

Type of soil	A (km ²)	Percentage (%)	K	A · K
Histosols	36.21	5.99	0.28	10.14
Inceptisols	246.44	40.74	0.23	56.68
Mollisols	150.55	24.89	0.20	30.11
Ultisols	171.71	28.39	0.16	27.47
Total	604.91	100.00	–	124.40
Composite K value	0.21			

Source: own study.

Krueng Keureuto River. The SDR value is important for estimating the rate at which sediment reaches the Krueng Keureuto River relative to the total sediment eroded in the watershed. Based on the calculation, the SDR value was 27% of the total eroded sediment reaching the river. Based on this value, the actual sediment was 3,526.32 Mg·y⁻¹.

The next factor considered was sediment inflow transported by river flow into the lake. This factor is directly related to the magnitude of the planned flood discharge and the sedimentation rate in the oxbow lake catchment area. The analysis shows that the amount of sediment entering the lake reached 136.87 Mg·y⁻¹. The conversion of sediment mass to volume was performed using volumetric weight analysis, yielding a value of 2.56 g·cm⁻³ from field-measured primary data. Based on this value, the annual sediment volume reached 53.47 m³·y⁻¹.

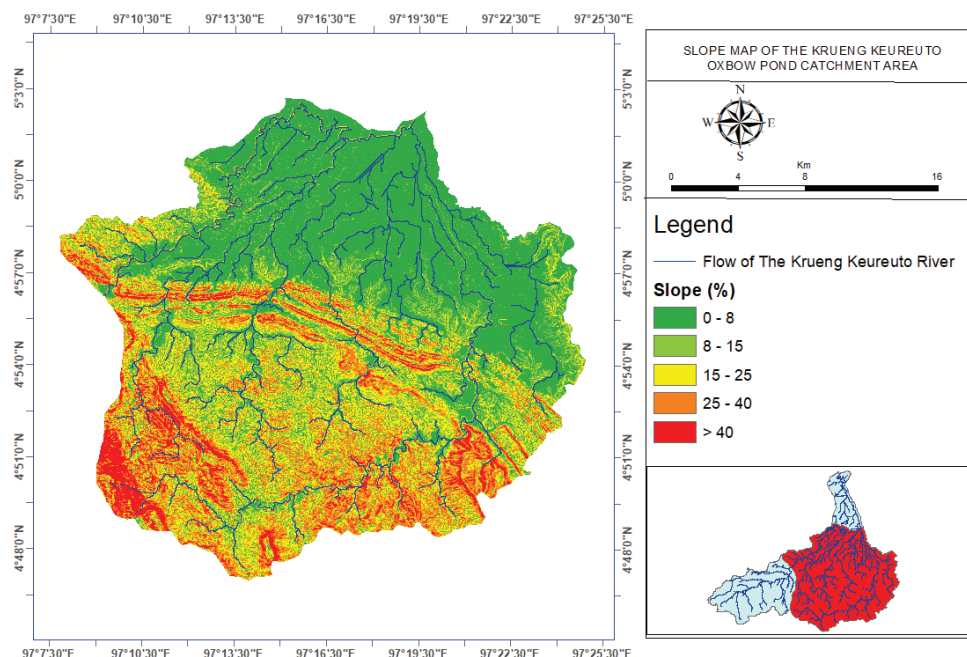


Fig. 3. Map of slope at Krueng Keureuto oxbow lake catchment area; source: own study

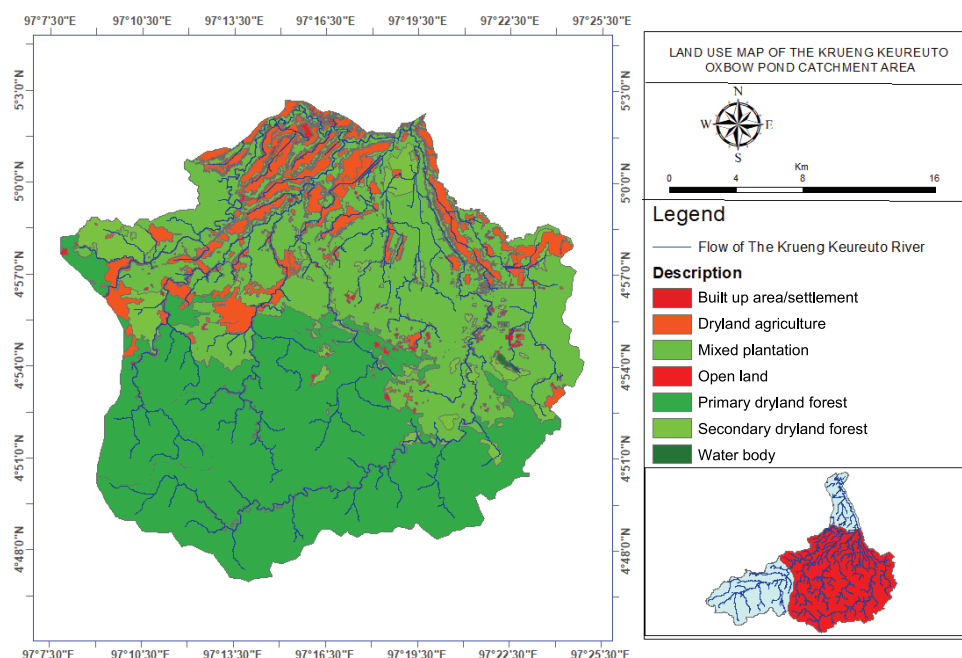


Fig. 4. Map of land use at the Krueng Keureuto oxbow lake catchment area; source: own study

Table 2. Land use values at the Krueng Keureuto oxbow lake catchment area

Land cover	A (km ²)	Percentage (%)	CP	A · CP
Water bodies (undefined)	4.18	0.69	0.00	0.00
Built land / settlement	9.69	1.60	0.95	9.21
Open land	3.87	0.64	1.00	3.87
Mixed plantation	217.99	36.04	0.50	108.99
Dryland agriculture	62.87	10.39	0.048	3.02
Secondary dryland forest	30.20	4.99	0.10	3.02
Primary dryland forest	276.12	45.65	0.0025	0.69
Total	604.91	100.00	–	128.80
Composite CP value			0.21	

Explanation: A = area, CP = crop and land conservation management factor.

Source: own study.

Table 3. Sedimentation calculation

Year	a	Q_p (m ³ ·s ⁻¹)	V_q (m ³)	b	K	LS	CP	SY (Mg·y ⁻¹)
2011	11.8	191.49	11,724.73	0.56	0.21	2.33	0.21	4,327.54
2012	11.8	97.42	8,196.73	0.56	0.21	2.33	0.21	2,425.60
2013	11.8	33.68	5,295.70	0.56	0.21	2.33	0.21	1,047.81
2014	11.8	1,277.73	43,540.79	0.56	0.21	2.33	0.21	26,117.54
2015	11.8	1,792.03	57,429.75	0.56	0.21	2.33	0.21	36,858.16
2016	11.8	140.57	9,875.72	0.56	0.21	2.33	0.21	3,306.09
2017	11.8	527.36	22,387.05	0.56	0.21	2.33	0.21	10,962.87
2018	11.8	410.78	18,872.81	0.56	0.21	2.33	0.21	8,662.29
2019	11.8	465.26	20,529.54	0.56	0.21	2.33	0.21	9,736.17
2020	11.8	858.86	31,942.33	0.56	0.21	2.33	0.21	17,578.60

cont. Tab. 3

Year	a	Q_p ($\text{m}^3 \cdot \text{s}^{-1}$)	V_q (m^3)	b	K	LS	CP	SY ($\text{Mg} \cdot \text{y}^{-1}$)
2021	11.8	1,639.26	53,331.64	0.56	0.21	2.33	0.21	33,640.20
2022	11.8	170.62	10,980.05	0.56	0.21	2.33	0.21	3,910.32
2023	11.8	521.74	22,220.04	0.56	0.21	2.33	0.21	10,851.66
Sedimentation rate ($\text{Mg} \cdot \text{y}^{-1}$)								13,032.68
Sedimentation rate ($\text{Mg} \cdot \text{ha}^{-1} \cdot \text{y}^{-1}$)								0.22

Explanations: a = MUSLE coefficient, Q_p = peak discharge, V_q = flow volume during an event, b = coefficient = 0.56, K = soil erodibility factor, LS = slope length and steepness, CP = crop and land conservation management factor, SY = sediment yield.

Source: own study.

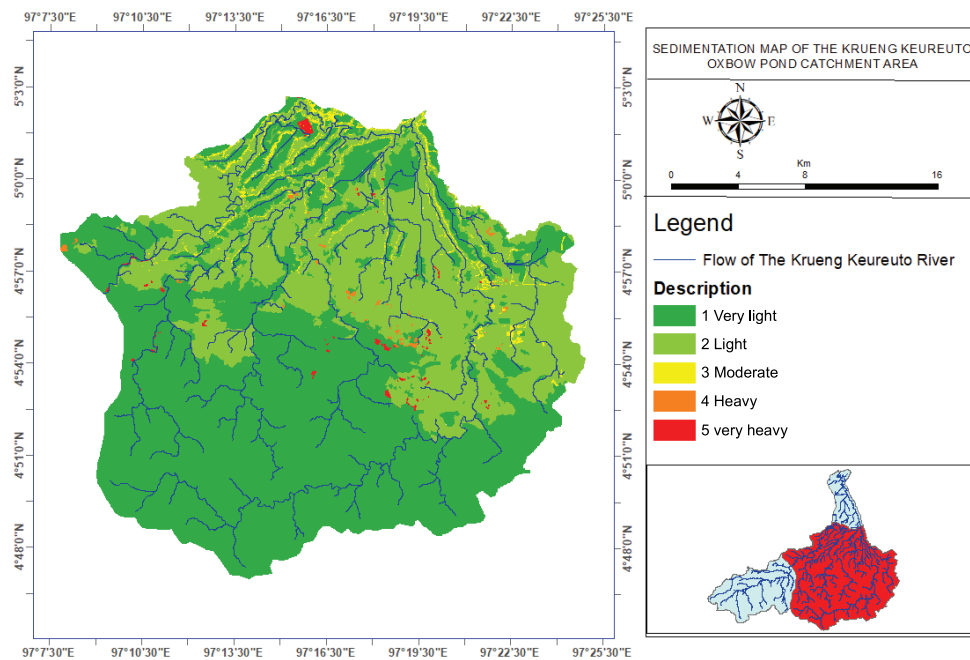


Fig. 5. Sedimentation map; source: own study

Table 4. Level of sedimentation at the oxbow lake catchment area

Sedimentation class	Area (km^2)	Percentage (%)
1	371.90	61.49
2	218.94	36.20
3	10.06	1.66
4	1.63	0.27
5	2.33	0.39
Amount	604.91	100.00

Source: own study.

DISCUSSION

SEDIMENT YIELD ANALYSIS

Determining oxbow basin boundaries using DEM and ArcGIS is appropriate for the relatively flat floodplain of the Krueng Keureuto watershed. However, the accuracy of watershed delineation strongly depends on DEM resolution and the correct

identification of hydrologically connected catchment areas. Accurate hydrological boundaries directly influence the estimation of peak discharge (Q_p), runoff volume (V_q), and sediment yield (SY) (Fatone *et al.*, 2021). Rainfall analysis for the 2011–2023 period from the Jambo Aye, Krueng Keureuto, and Panton Labu stations indicates high temporal variability, with a maximum annual rainfall of 184.70 mm in 2015 and a minimum of 59.11 mm in 2013. These variations reflect the influence of monsoon variability and ENSO-related climate anomalies in northern Aceh. Years with higher rainfall intensity, particularly 2014–2015 and 2020–2021, correspond to increased runoff and erosion processes, consistent with findings by Anbazhagan, Nanditha and Reshimidevi (2024) and González *et al.* (2022), who reported that rainfall intensity and temporal distribution significantly affect runoff generation and sediment transport.

The Soil Conservation Service (SCS) hydrological analysis produced peak discharge values ranging from 33.68 to 1,792.03 $\text{m}^3 \cdot \text{s}^{-1}$, with an average of approximately 643 $\text{m}^3 \cdot \text{s}^{-1}$. The highest peak discharge occurred in 2015 and reflected a rapid hydrological response to high-intensity rainfall events. Runoff volumes ranged from 5,295.70 m^3 to 57,429.75 m^3 , indicating a transition from infiltration-controlled conditions in dry years to

runoff-dominated conditions in wet years. Such nonlinear hydrological responses are common in tropical catchments experiencing high rainfall variability (Reda, Moges and Kendie, 2024; Wang and Bi, 2020). Increased Q_p and V_q directly enhance the energy of surface runoff and consequently increase sediment transport toward the oxbow lake (Zantet Oybitet, Bibi and Adem, 2023). Similar findings were reported by Gusma *et al.* (2023), who demonstrated that watershed characteristics, rainfall, land use, and topographic conditions strongly influence soil erosion rates and sediment hazard levels. Their study further highlights the importance of erosion assessment as a basis for sediment management and the long-term sustainability of water storage infrastructure.

Soil characteristics also play a significant role in sediment generation. The dominant soil types in the watershed are inceptisols (40.74%), followed by ultisols (28.39%), mollisols (24.89%), and histosols (5.99%). Histosols show the highest erodibility value ($K = 0.28$), while ultisols exhibit the lowest value ($K = 0.16$) due to their stable clay structure. Spatial analysis using ArcGIS 10.8 indicates that areas with relatively high K values (≥ 0.2) are concentrated in upstream and midstream regions. According to Komissarov *et al.* (2024), soils with stable fine structures are generally more resistant to detachment, whereas soils with higher sand fractions tend to show greater erodibility.

The composite soil erodibility factor (K) for the watershed is estimated at 0.21, indicating moderate susceptibility to erosion. Inceptisols contribute the largest area (246.44 km²), and their silty clay texture facilitates the formation of fine sediment particles that can be transported during high rainfall events. Previous studies have shown that variations in soil erodibility can significantly influence sediment yield estimates in modified universal soil loss equation (MUSLE)-based simulations (Krajewski, Banasik and Hejduk, 2024).

The integration of slope length and steepness (LS) and crop and land conservation management factor (CP) spatial factors shows that areas with $LS > 1$ and $CP > 0.3$ represent priority zones for soil conservation because they potentially generate 60–80% of the total sediment load (Safira *et al.*, 2023; Zantet Oybitet, Bibi and Adem, 2023). Vegetative conservation measures such as riparian buffers and slope stabilisation can reduce sediment yield by up to 35% (Wang *et al.*, 2023). Overall, the composite CP value of 0.21 suggests that the watershed is moderately protected against erosion, although increasing anthropogenic activities in open land and plantation areas may elevate sediment production.

Despite its practical applicability, the MUSLE approach contains several inherent limitations that should be considered when interpreting the results. The model estimates sediment yield primarily from runoff energy and watershed characteristics, but it does not explicitly represent complex sediment transport processes such as channel erosion, intermediate deposition, or temporal variations in sediment connectivity within the watershed. Consequently, the model may simplify spatial variability of sediment sources and transport pathways (Reda, Moges and Kendie, 2024).

In addition, the accuracy of MUSLE predictions depends strongly on hydrological inputs such as peak discharge and runoff volume. Uncertainty in these parameters can propagate through the model and influence sediment yield estimates (Tsige, Malcherek and Salesi, 2022). Therefore, the sediment yield

calculated in this study should be interpreted as a reasonable approximation rather than an exact measurement, particularly in ungauged or data-limited watersheds.

THE LIFESPAN OF KRUENG KEUREUTO OXBOW LAKE

The storage capacity of an oxbow lake represents its ability to retain water and sediment, consisting of effective storage (live storage) and dead storage. Live storage supports operational functions such as irrigation and flood control, while dead storage acts as a sediment reservoir during the lake's operational life. In this study, the estimation of oxbow lake lifespan is based on the dead storage capacity derived from the elevation–volume relationship. According to Harahap (2025), the dead storage capacity at an elevation of +1 m is 20,348.57 m³.

Trap efficiency represents the ability of the lake to retain incoming sediment and is estimated using the Brune method, which relates storage capacity (C) to annual inflow (I) (Sisinggih *et al.*, 2023).

Based on research by Harahap (2025), the horseshoe-shaped pond in the Krueng Keureuto watershed has a storage capacity of 909,228 m³ and can accommodate an inflow (Q_{inflow}) of 257.88 m³·s⁻¹ or approximately 8,132,503,680 m³·y⁻¹ during flood events between the 7th and 11th hours. After this period, the river water level falls below the spillway threshold, and inflow into the pond stops. From the 11th hour onwards, the flow is characterised as $Q_{outflow}$, indicating the emptying phase. Under these conditions, the discharge entering during the Q_{inflow} phase is retained within the pond, allowing transported sediment to settle. Therefore, the C/I ratio is assumed to be equal to one.

The estimated operational lifespan of the Krueng Keureuto oxbow lake is approximately 3.9 years, indicating that sediment accumulation may saturate the dead storage capacity within a relatively short period. Sediment trapping efficiency can vary significantly during flood events, and higher sediment retention rates accelerate the reduction of available storage capacity (Liu *et al.*, 2025). This finding aligns with Sisinggih *et al.* (2023), who emphasised that trap efficiency is a key parameter in assessing reservoir sedimentation and storage loss.

Trap efficiency curves are generally derived from observations in large reservoirs, and their application to smaller floodplain water bodies such as oxbow lakes may introduce uncertainty (Sisinggih *et al.*, 2023). Sediment deposition in oxbow systems is influenced by floodplain morphology, hydraulic connectivity with the main river, and variable flood dynamics. Consequently, the lifespan estimated using the Brune method should be interpreted as a first-order approximation rather than a precise prediction of long-term lake performance (Ren *et al.*, 2024).

The estimated lifespan of approximately 3.9 years should therefore be interpreted as a baseline condition representing current sediment inflow without additional sediment management. In tropical watersheds with high rainfall intensity, rapid sediment accumulation often occurs when upstream erosion remains uncontrolled (Asres *et al.*, 2025; Liu *et al.*, 2025). From a flood mitigation perspective, oxbow lakes can still contribute to peak flow attenuation, but their long-term effectiveness depends on sediment management measures such as upstream erosion control, riparian buffer implementation, and periodic dredging (Robotham *et al.*, 2021).

In cyclic oxbow systems, sediment inflow is strongly influenced by seasonal flooding. The filling and emptying processes occur only when river discharge exceeds a certain threshold. Consequently, sediment accumulation increases during years with frequent flooding. Therefore, continuous sediment management and periodic dredging are necessary to maintain storage capacity and sustain the functional role of oxbow lakes in flood mitigation.

CONCLUSIONS

This study shows that the Krueng Keureuto oxbow lake plays an important role in storing flood runoff while retaining sediment carried by the flow. The application of the modified universal soil loss equation (MUSLE) method revealed that the watershed is dominated by inceptisols, with primary dryland forest as the largest land cover type. The slope length and steepness (*LS*) value was 2.33, contributing to the high annual sedimentation rate of $53.47 \text{ m}^3 \cdot \text{y}^{-1}$ during the period 2011–2023. Based on the lake's dead storage capacity and trap efficiency, the oxbow lake has an operational life of approximately 3.9 years until the dead storage is filled to capacity.

This relatively short operational lifespan reflects the current unmanaged condition of the oxbow lake, where no sediment control or maintenance measures are implemented. The findings highlight the importance of sediment management and regular maintenance planning to maintain the lake's function as flood-control infrastructure in the Krueng Keureuto watershed.

The main scientific contribution of this study lies in providing a quantitative framework for estimating sediment accumulation and evaluating the operational lifespan of natural oxbow lakes using an integrated MUSLE and trap efficiency approach. This framework improves the understanding of sediment dynamics in tropical floodplain environments where oxbow lakes are periodically connected to the main river during flood events.

From a broader perspective, the results emphasise the importance of managing sedimentation to sustain the role of oxbow lakes as nature-based flood mitigation systems. The methodological approach applied in this study may also be transferable to other floodplain oxbow lakes in tropical watersheds facing similar sedimentation challenges. Future research is recommended to incorporate hydrological variability, flood connectivity dynamics, and watershed management scenarios to further evaluate the long-term sustainability of oxbow lakes as nature-based solutions for flood mitigation.

SUPPLEMENTARY MATERIAL

Supplementary material to this article can be found online at https://www.jwld.pl/files/Supplementary_material_70_Azmeri.pdf.

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CONFLICT OF INTERESTS

All authors declare that they have no conflict of interests.

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